

Verilog HDL – SR FF

```
module srff (s,r,clk,rst,q,qbar);  
    input s,r,clk,rst;  
    output q,qbar;  
    reg q;  
    always@(negedge clk)  
    begin  
        if (rst)  
            q <= 1'b0;  
        else if (s==1'b0 && r==1'b0) q <= q; // Hold  
        else if (s==1'b0 && r==1'b1) q <= 1'b0; //Reset  
        else if (s==1'b1 && r==1'b0) q <= 1'b1; //Set  
        else if (s==1'b1 && r==1'b1) q <= 1'bz; //Invalid  
    end  
    assign qbar = ~q;  
endmodule
```

SR Flipflop Testbench

```
module srfftb();
    reg r,s,clk,rst;
    wire q;
    srff sr1(s,r,clk,rst,q,qbar);
    initial
    begin
        clk=1'b0;  rst=1'b1; s=1'b0; r=1'b1;
    end
    always
        #100 clk=~clk;
    initial
    begin
        #200 rst=1'b0;
        #200 s=1'b1;r=1'b0;
        #200 s=1'b0;r=1'b0;
        #200 s=1'b1;r=1'b1;
        #200 rst=1'b1;
    end
    initial
    $monitor($time,"s=%b,r=%b,q=%b,rst=%b",s,r,q,rst);
endmodule
```

Verilog HDL - DFF

```
module dff (d,clk,rst,q,qbar);  
  input d,clk,rst;  
  output q,qbar;  
  reg q;  
  always@(posedge clk)  
  begin  
    if (rst)  
      q <= 1'b0;  
      /* else if (d == 1'b0)  
        q <= 1'b0;  
      else  
        q <= 1'b1; */  
    else  
      q <= d;  
    end  
  assign qbar = ~q;  
endmodule
```

D Flip flop Testbench

```
module dfftb();
    reg d,rst,clk;
    wire q;
    dff d1 (d,clk,rst,q,qbar);
    initial
    begin
        clk=1'b0; rst=1'b1; d=1'b0;
    end
    always
        #100 clk=~clk;
    initial
    begin
        #200 rst=1'b0;
        #200 d=1'b1;
        #200 d=1'b0;
        #600 rst=1'b1;
    end
    initial
    $monitor($time,"q=%b,d=%b,rst=%b",q,d,rst);
endmodule
```